

Perception with Zed X

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Introduction to Perception

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Camera Specifications

There are 3 steps to digital camera functions:

1. integration
2. collection
3. reading
4. amplification
5. imaging
6. buffering (optional)

In integration, light must reach an array of photodiodes to (generate a charge?) these diodes are slightly more sensitive to the infrared end of the visible light spectrum having a range of 400-1000 nm. Then the reading process begins by a collection of charges from the photodiodes read by a specialized chip that varies

on the model of camera; charge coupled device (CCD) or complementary metal oxide on silicon (CMOS). Finally, the signals are amplified and sent out through an imaging chip (Siegwart et al., 2011, p. 143).

Camera configurations change the result of this process. For example, the duration of this period and amount of light exposed to the diodes are critical to the results; it's important to regulate the amount of photons. When too much light is processed in the collection step this causes blooming where the saturation of the light in the collection causes light to leak to adjacent pixels in its array. modifying the integration and collection This can be done by changing the aperture (iris position) and shutter speed to modify the integration phase and collection phase respectively (Siegwart et al., 2011, p.143-145)

Zed X

The Zed X is an environmentally sealed stereo camera using complementary metal oxide semiconductor chip (CMOS) sensors¹². As shown in Table 1, the ZedX CMOS has an array size of 1928 x 1208 photodiodes.

¹StereoLabs, 2023.

²“Neousys Enhances GMSL2 Vision with Stereolabs ZED X and ZED X One GS Cameras”, 2025, paragraph 1.

Table 1: Zed X Camera Specifications

Sensor Type	CMOS ^a
Array Size	1928 x 1208 px
Pixel Size	3 x 3 micrometers
Shutter	Electronic Synchronized global shutter
Output Format	RAW10
Max SN Ratio	38 dB
Dynamic Range	71.4dB
Responsivity	22.3KeLux-sec
Baseline	12cm

^aData from technical specification (StereoLabs, 2023).

^aSpecifically retrieved from Neousys Technology article (“Neousys Enhances GMSL2 Vision with Stereolabs ZED X and ZED X One GS Cameras”, 2025).